# 1.Tic Tac Toe

Welcome | tictactoe.py | simplevacuum.py | goalvacuum.py

tictactoe.py > ...

```python
board = [" "] * 9

def display():
    print(board[0], "|", board[1], "|", board[2])
    print("--+---+--")
    print(board[3], "|", board[4], "|", board[5])
    print("--+---+--")
    print(board[6], "|", board[7], "|", board[8])

def check():
    win = [(0,1,2), (3,4,5), (6,7,8),
           (0,3,6), (1,4,7), (2,5,8),
           (0,4,8), (2,4,6)]

    for a, b, c in win:
        if board[a] == board[b] == board[c] != " ":
            return True
    return False

player = "X"

for i in range(9):
    display()
    pos = int(input("Player " + player + " position (1-9): ")) - 1

    if board[pos] == " ":
        board[pos] = player

        if check():
            display()
            print(player, "wins!")
            break

        player = "O" if player == "X" else "X"
else:
    display()
    print("Draw!")
```

```
PS C:\Users\BMSCE\Desktop\AI lab> c:\python314\python.exe "c:/Users/BMSCE/Desktop/AI lab/tictactoe.py"
  |   |
Player O position (1-9): 3
  |   | O
--+---+--
  | X |
--+---+--
  |   |
Player X position (1-9): 2
  | X | O
--+---+--
  | X |
--+---+--
  |   |
Player O position (1-9): 4
  | X | O
--+---+--
O | X |
--+---+--
  |   |
Player X position (1-9): 6
  | X | O
--+---+--
O | X | X
--+---+--
  |   |
Player O position (1-9): 7
  | X | O
--+---+--
O | X | X
--+---+--
O |   |
Player X position (1-9): 8
  | X | O
--+---+--
O | X | X
--+---+--
O | X |
X wins!
PS C:\Users\BMSCE\Desktop\AI lab>
```

# 2. Goal Based Vacuum

Welcome | tictactoe.py | simplevacuum.py | goalvacuum.py

goalvacuum.py > ...

```python
def goal_based_agent(location, rooms):
        return "Suck"

    if rooms["A"] == "Dirty":
        if location == "A":
            return "Suck"
        else:
            return "Move Left"

    if rooms["B"] == "Dirty":
        if location == "B":
            return "Suck"
        else:
            return "Move Right"

    return "Stop"

location = "A"

rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

print("GOAL-BASED VACUUM CLEANER")
print("------------------------")

while True:
    print(f"\nLocation : {location}")
    print(f"Room A   : {rooms['A']}")
    print(f"Room B   : {rooms['B']}")

    action = goal_based_agent(location, rooms)
    print(f"Action   : {action}")

    if action == "Suck":
        rooms[location] = "Clean"

    elif action == "Move Right":
        location = "B"

    elif action == "Move Left":
        location = "A"
```

Problems Debug Console Terminal Ports

```
PS C:\Users\BMSCE\Desktop\AI lab> & c:\python314\python.exe "c:/Users/BMSCE/Desktop/AI lab/goalvacuum.py"
GOAL-BASED VACUUM CLEANER
-------------------------

Location : A
Room A   : Dirty
Room B   : Dirty
Action   : Suck

Location : A
Room A   : Clean
Room B   : Dirty
Action   : Move Right

Location : B
Room A   : Clean
Room B   : Dirty
Action   : Suck

Location : B
Room A   : Clean
Room B   : Clean
Action   : Stop

Goal achieved: Both rooms are clean!
PS C:\Users\BMSCE\Desktop\AI lab>
```

# 3. Simple Vacuum

Welcome | tictactoe.py | simplevacuum.py | goalvacuum.py

simplevacuum.py > ...

```python
def simple_reflex_agent(location, status):
    if status == "Dirty":
        return "Suck"

    if status == "Clean":
        if location == "A":
            return "Move Right"
        else:
            return "Move Left"


location = "A"
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

print("SIMPLE REFLEX VACUUM CLEANER")
print("----------------------------")

for step in range(6):
    status = rooms[location]

    print(f"\nLocation : {location}")
    print(f"Status   : {status}")

    action = simple_reflex_agent(location, status)
    print(f"Action   : {action}")

    if action == "Suck":
        rooms[location] = "Clean"

    elif action == "Move Right":
        location = "B"

    elif action == "Move Left":
        location = "A"

    if rooms["A"] == "Clean" and rooms["B"] == "Clean":
        print("\nBoth rooms are clean.")
        break

```

Problems Debug Console **Terminal** Ports

```
PS C:\Users\BMSCE\Desktop\AI lab> & c:\python314\python.exe "c:/Users/BMSCE/Desktop/AI lab/simplevacuum.py"
SIMPLE REFLEX VACUUM CLEANER
----------------------------

Location : A
Status   : Dirty
Action   : Suck

Location : A
Status   : Clean
Action   : Move Right

Location : B
Status   : Dirty
Action   : Suck

Both rooms are clean.
PS C:\Users\BMSCE\Desktop\AI lab>
```